

Generative AI Can Improve Performance and Engagement without Harming Learning

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Abstract

We study how the usage of generative AI shapes both students' performance and long-run skill development within a large-scale field setting at a major EdTech platform. A growing literature documents that generative AI can help users complete tasks faster and more accurately when having access to the tool, but may crowd out individual effort and undermine skill development when used during problem-solving. In this paper, we examine a specific design in which AI is activated only *after* students submit an answer and is used exclusively for post-solution debriefing: analyzing mistakes, explaining solutions, and answering follow-up questions. Leveraging the partial rollout of an AI tutor across thousands of learning centers, we analyze a random panel of 19,709 students (over 2.2 million student-day observations) before and after the introduction of generative AI. A difference-in-differences design reveals that access to the AI tutor substantially increases student engagement: treated students complete 36% more math problems. We also find improvements in learning efficiency, as students spend 3.9% less time per problem while their correctness ratio increases by 3.6%. Heterogeneity analyses show that the AI tutor disproportionately benefits lower-performing students and, at the same time, does *not* generate larger gains for students from higher-income areas, suggesting that the technology narrows performance gaps without widening income-based disparities. Finally, we find evidence of long-run skill development, as students with post-solution AI support convert study time into improved performance at higher rates than those without AI access.

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1 Introduction

Artificial intelligence (AI) is increasingly seen as a transformative force in education, promising to deliver cost-effective, high-quality learning at scale. By providing personalized instruction, immediate feedback, and adaptive learning paths, AI-based systems have the potential to address longstanding challenges in educational productivity and effectiveness. In fact, meta-analyses of intelligent tutoring systems and large-scale experiments with technology-aided instruction show that well-designed digital tools can generate substantial gains in test scores, often particularly for students who start out behind (Banerjee et al., 2007; Steenbergen-Hu and Cooper, 2013; Ma et al., 2014; Muralidharan, Singh, and Ganimian, 2019). In theory, generative AI can further improve educational effectiveness. Yet despite this optimism, emerging evidence shows that generative AI can *harm* learning when used as an unstructured assistant during problem solving rather than as a carefully designed tutoring tool (Urban et al., 2024; Choi et al., 2021). In a recent large-scale randomized experiment, high school students given unrestricted access to a ChatGPT-style assistant during mathematics practice performed significantly worse on subsequent unassisted exams, indicating that reliance on generative AI crowded out the effortful cognitive processes that drive long-term learning (Bastani et al., 2025). Complementary meta-analytic evidence on ChatGPT in education finds that, while average performance effects are often positive, there are persistent concerns that students may offload too much cognitive work to the model, reducing mental effort and potentially undermining deeper learning (Deng et al., 2024). These results echo broader research in cognitive science showing that minimal guidance and heavy reliance on external aids can impede schema formation and long-term skill development (Kirschner, Sweller, and Clark, 2006).

In this paper, we argue that the design of generative AI, particularly its role within the learning workflow, is critical for determining its impact on learning. Rather than viewing AI as uniformly beneficial or harmful, our results, together with existing evidence, indicate that its effect may depend on the timing and nature of its intervention. Prior work shows that using AI during problem solving, for example, to obtain immediate answers or to complete intermediate steps, can impede the learning process (Kirschner, Sweller, and Clark, 2006; Bastani et al., 2025). By contrast, we study a setting in which AI operates strictly as a post-solution tool: it is activated only after the student submits an answer and is used exclusively for debriefing, including analyzing mistakes, explaining solutions, and answering follow-up questions. This design preserves the student’s opportunity to engage independently with the problem and then uses AI to convert errors into targeted learning

moments. Consistent with the literature on effective feedback, which emphasizes the importance of timely, task-focused information that helps learners close the gap between current performance and desired goals (Hattie and Timperley, 2007), we show that this post-solution configuration yields sizable improvements in engagement, efficiency (time per question), correctness, and longer-run learning, without evidence of the complacency or dependency effects that may arise when AI is available during problem solving from the very beginning. By confining AI to the debriefing phase of the learning cycle, students benefit from personalized feedback and on-demand clarification while avoiding substitution for the student’s own problem-solving effort.

Another crucial consideration is equity in educational outcomes and the central question of whether AI-enabled learning tools narrow or widen existing gaps. On one hand, AI may amplify inequality if higher-performing students are better positioned to adopt and leverage such tools because they have stronger baseline skills. On the other hand, AI’s ability to deliver personalized feedback and adapt to individual learning trajectories provides a mechanism through which lower-performing students may catch up. Evidence from earlier waves of technology-aided instruction shows that personalized, level-appropriate content can generate especially large gains for academically weaker students, thereby reducing achievement gaps (Banerjee et al., 2007; Muralidharan, Singh, and Ganimian, 2019). Related evidence from professional writing and customer support contexts indicates that generative AI assistants have been shown to raise productivity and quality while disproportionately helping lower-skilled workers, leading to a compression of performance differences (Noy and Zhang, 2023; Dell’Acqua et al., 2023; Brynjolfsson, Li, and Raymond, 2025).

In our setting, this mechanism is particularly salient: the AI tutor is activated only after a solution is submitted and engages more intensively when a mistake is made. As a result, students who make more errors naturally receive more AI feedback and explanation. This design implies that lower-performing students, who make errors more frequently, interact more with the AI tutor and may therefore experience larger learning gains, which in turn can close performance gaps. Consistent with this mechanism, we find that the benefits of the AI tutor are especially pronounced for students who start with lower baseline performance. These initially low-performing students exhibit larger improvements in correctness rate and efficiency relative to higher-performing peers, so that the performance gap between low- and high-performing students shrinks after the introduction of post-solution AI tutoring. Moreover, using local income levels as a proxy for socioeconomic status, we find no evidence that students from higher-income areas derive disproportionate benefits. Students from lower- and middle-income locations experience comparable or slightly larger gains

across several outcomes. In this online EdTech setting, AI adoption therefore tends to narrow performance gaps without exacerbating income-based disparities.

Our results also shed light on the mechanisms through which post-solution generative AI affects learning. First, we find that access to the generative AI tutor meaningfully increases engagement with the platform: treated students are active on more days per week, and conditional on being active, they solve more problems and devote more time to problem solving on the education platform. A natural interpretation is that timely, targeted resolution of confusion after an error is central to sustaining motivation. When students make mistakes but do not receive clear, prompt explanations, confusion can accumulate, leading to frustration and disengagement. In our setting, the post-solution AI tutor steps in precisely at this critical moment, providing an immediate debrief, clarifying the source of the error, and answering follow-up questions. By turning mistakes into structured learning opportunities rather than unresolved dead ends, the system appears to make continued practice more appealing and thus increases students’ willingness to return and invest additional time. Second, we find that each question takes less time to complete for students with AI access, even as overall engagement rises. This pattern is consistent with the AI tutor reducing “frictional” time that would otherwise be spent circling around unresolved misunderstandings. Instead of lingering in confusion after an error or searching for external help, students can quickly obtain an explanation and move on to the next problem with a clearer mental model. Taken together, these findings suggest a complementary pair of mechanisms: post-solution generative AI both raises the quantity of practice (by sustaining motivation and platform use) and improves the efficiency of that practice (by resolving confusion more quickly), thereby amplifying its positive impact on learning outcomes.

Our results so far indicate the AI tutor is most effective when it builds on a student’s own attempt, helping to interpret mistakes and solidify concepts rather than replacing the underlying cognitive work. This naturally raises the question of whether very frequent use of the AI tutor carries potential downsides. While post-solution debriefs clearly add value, there may be some situations in which conducting at least part of the debrief independently—such as trying to understand an error before asking the AI, or occasionally working without AI support—proves more beneficial, precisely because the ideal relationship is one of complementarity rather than pure substitution. Our empirical findings are consistent with this interpretation. We find evidence of diminishing marginal returns to AI usage: the immediate performance gains from the generative AI tutor diminish when the tool has been used very intensively in the recent past. Similarly, the

effect of the AI tutor on performance is larger, following a longer gap since the last AI-supported session. This pattern suggests that students learn more effectively when AI support is interspersed with periods of independent problem solving, rather than being invoked mechanically after every attempt. Taken together, these results reinforce the view that generative AI is most powerful when it complements, rather than crowds out, students’ own learning efforts, and they highlight the importance of designing usage norms and product interfaces that encourage this complementary mode of interaction.

Finally, we speak directly to concerns about long-term skill development under generative AI. In the short run, our design is intended to complement, rather than replace, students’ independent thinking: generative AI is introduced only at the debriefing stage, after the student has attempted the problem and submitted an answer. Because the tool does not generate solutions *ex ante*, it preserves the need for students to exert effort, formulate an approach, and commit to an answer before receiving assistance. Nonetheless, even in this post-solution configuration, one might worry that repeatedly providing AI-generated debriefs and explanations could gradually reduce students’ intrinsic effort or reliance on their own reasoning, thereby impeding longer-term skill formation. To assess this concern, we examine how students’ correctness rate evolves as they accumulate study time and practice, and we find that learners with access to post-solution generative AI tutoring show steeper trajectory of improvements in correctness rate than comparable students without access. In other words, conditional on the same amount of practice, treated students convert study time into accuracy gains at a higher rate. This pattern is at odds with the view that post-solution AI feedback crowds out learning; instead, it suggests that generative AI, when confined to the debriefing stage, can accelerate the acquisition of durable skills. Rather than merely boosting short-term performance, our results indicate that post-solution AI tutoring supports faster improvement in underlying competence over time, in line with theories of learning that emphasize the role of timely, information-rich feedback in promoting skill accumulation ([Hattie and Timperley, 2007](#)).

Taken together, our results suggest that generative AI, when limited to post-solution debriefing, can motivate students to work harder and improve their correctness at a faster rate. It also helps in leveling the playing field across different performance and socioeconomic levels, without much evidence of negative long-term effects on skill acquisition. Given that this approach is relatively cheap (discussed further in the data description) and highly scalable, it has great potential for educators and policymakers to use to improve educational outcomes, as well as for possible applications in other domains where structured feedback and reflection are critical.

The rest of the paper is organized as follows. Section 2 summarizes the related literature and outlines our contributions. Section 3 describes the data and field setting. Section 4 analyzes the main effects of the AI tutor mode. Section 5 examines heterogeneous effects and explores potential mechanisms. Section 6 presents additional analyses on learning dynamics. Section 7 concludes.

2 Related Literature

Our study connects three broad streams of research: (i) technology-mediated and online learning, (ii) human-AI collaboration, and (iii) the cognitive and distributional consequences of AI-based tutoring and personalization. We position our contribution at the intersection of these literatures by examining how a post-solution generative AI tutor, embedded in a large commercial EdTech platform, affects engagement, learning efficiency, long-run skill development, and inequality in learning outcomes.

2.1 Technology-Mediated and Online Learning

The existing literature, especially within the information systems field, has long emphasized that digital technologies can fundamentally reshape how individuals learn and acquire skills and knowledge. Early work documented how electronic classrooms and networked computing change the organization of instruction and patterns of student interaction (Leidner and Jarvenpaa, 1993). Alavi and Leidner (2001) argued that technology-mediated learning (TML) should be studied not merely as a delivery mechanism, but as a socio-technical system that reconfigures pedagogy, learner roles, and institutional arrangements. Related studies examined technology-supported training and skill acquisition, showing that IT-based instruction interacts with learners’ beliefs, self-efficacy, and mental models (Compeau and Higgins, 1995; Vandenbosch and Higgins, 1996; Yi and Davis, 2003).

Within this stream, a large body of work has examined how different designs of technology-mediated learning environments shape engagement, satisfaction, and achievement. Alavi (1994, 1995) show that computer-mediated collaborative learning and distributed telelearning can improve higher-order skills and perceptions of learning quality relative to traditional classroom arrangements. Subsequent research underscored the importance of instructional design and learner self-regulation in e-learning contexts. Santhanam, Sasidharan, and Webster (2008) demonstrate that supporting self-regulatory learning strategies substantially enhances the effectiveness of e-learning-based IT training, while Santhanam, Liu, and Shen (2016) show that gamification of

technology-mediated training can improve learning outcomes, but only when competitive elements are carefully designed. Complementary work argues that research on learning must explicitly consider realism, relevance, and rigor, and view technology as embedded in broader learning processes and contexts (Gupta and Bostrom, 2013).

More recent work has shifted toward large-scale digital platforms and fine-grained experimentation. In a series of field experiments on an online learning platform, Huang et al. (2021) show that simple commitment devices and reminders can combat procrastination and significantly increase course completion. Leung et al. (2023) study gamification designs in an online learning environment and find that personalized gamification can enhance engagement, but its effectiveness depends on aligning game elements with learner characteristics. In the context of course and content design, Li et al. (2024) show that interleaved e-learning designs—which algorithmically mix topics rather than block them—can improve retention and performance relative to traditional blocked designs, illustrating how data-driven design choices translate cognitive science insights into practice.

Our setting is closely related conceptually, but differs in two important dimensions. First, we study a commercial K–12 math learning platform in which students repeatedly engage over months, generating millions of student–day observations. This scale allows us to analyze not only engagement and immediate correctness but also how learning curves evolve as students accumulate practice. Second, we focus on a *post-solution generative AI tutor* that intervenes only after a student submits an answer. In contrast to most prior TML work, which centers on static content delivery, gamified progress cues, or rule-based tutoring, our context involves a generative AI system that analyzes student solutions, provides step-by-step debriefs, and answers follow-up questions. Our results show that this specific placement of AI within the learning workflow meaningfully increases platform engagement, improves learning efficiency, and steepens learning curves, thereby extending the TML literature into the era of large language model (LLM)–driven tutoring.

2.2 Human–AI Collaboration and Algorithmic Decision Support

A second related stream examines how algorithms and AI systems interact with human decision makers. Early research on decision support systems (DSS) established that well-designed DSS can improve decision processes and outcomes by providing models, data, and structured analyses to human users (Alavi, 1981; Sharda, Barr, and McDonnell, 1988; van Bruggen, Smidts, and Wierenga, 1998). These studies emphasized that the value of DSS depends on fit with user tasks, decision

processes, and organizational context, anticipating later work on task–technology fit more broadly (Goodhue and Thompson, 1995).

As data-rich digital environments emerged, research began to examine algorithmic recommendations and online platforms. Aral and Walker (2011) and Bapna and Umyarov (2015) use large-scale field experiments in social networks to show how algorithmically mediated peer signals affect adoption and willingness to pay, highlighting the subtle ways in which algorithmic systems shape human behavior. Related work has investigated the impact of personalization and recommendation agents on user preferences and post-consumption evaluations (Bauman and Tuzhilin, 2018; Adomavicius et al., 2022), documenting that algorithms do not merely reflect preferences but can also shift them.

A central insight of the more recent human-AI collaboration literature is that humans often exhibit nuanced patterns of trust, reliance, and resistance toward algorithmic advice. Dietvorst, Simmons, and Massey (2018) show that individuals are reluctant to rely on algorithms, particularly after observing them make errors, but that allowing users even limited control over algorithmic outputs can substantially increase adoption. Gregor and Benbasat (1999) and Komiak and Benbasat (2006) demonstrate that explanation facilities, personalization, and familiarity with intelligent agents can improve understanding and trust, which in turn shape usage intentions. More broadly, Berente et al. (2021) discuss how AI systems restructure organizational roles and routines, while Lebovitz, Levina, and Lifshitz-Assaf (2021) highlight the dangers of treating AI-generated “ground truth” as objective, emphasizing the need for human oversight and reflexivity.

Recent work has also examined performance implications of human–AI collaboration in high-stakes domains. Deng et al. (2022) study an AI-powered shelf monitoring system in retail and find that AI-assisted teams increase product sales; interestingly, some benefits persist even after the AI is removed, suggesting that workers learn from AI feedback. Abdel-Karim et al. (2023) analyze AI-augmented diagnostic work in radiology and show that AI systems can induce productive reflection when they occasionally disagree with human experts, improving calibration and diagnostic accuracy over time. At the same time, Turel and Kalhan (2023) document implicit biases against machine advice that can lead to under-utilization of AI, particularly early in adoption.

Our study contributes to this literature in three ways. First, we examine human–AI collaboration not in a single decision task but in a repeated learning environment, where students iteratively solve problems, receive AI feedback, and update their knowledge. Second, the AI tutor in our setting is deliberately positioned as a *post-solution* decision support tool: it does not generate answers *ex ante* but instead analyzes completed solutions and provides explanations afterward.

This design reflects principles from the algorithm-aversion and explanation literatures—maintaining human effort upfront while using AI to clarify and correct *ex post*. Third, we provide large-scale field evidence on how such a design affects downstream performance and trajectories of skill development, thereby extending human-AI collaboration research from decision quality to long-run learning outcomes.

2.3 Cognitive and Distributional Effects of AI in Learning

A third stream investigates how technology and AI shape cognitive processes and the distribution of learning outcomes. The existing work has long emphasized that active engagement, collaborative problem solving, and feedback are crucial for deep learning (Leidner and Jarvenpaa, 1993; Alavi, 1994, 1995; Alavi and Leidner, 2001). Empirical studies of technology-mediated learning show that designs which encourage self-regulation, reflection, and deliberate practice tend to produce more durable improvements in skill (Compeau and Higgins, 1995; Santhanam, Sasidharan, and Webster, 2008; Gupta and Bostrom, 2013; Santhanam, Liu, and Shen, 2016). In this view, tools that overly simplify tasks risk reducing opportunities for “productive struggle”, potentially weakening long-run skill formation.

Emerging evidence on generative AI in education echoes these concerns. Recent randomized and quasi-experimental studies show that unrestricted access to ChatGPT-style tools during coursework can reduce subsequent unaided performance when students rely on AI to generate solutions rather than using it as a scaffold for learning (Choi et al., 2021; Urban et al., 2024; Bastani et al., 2025). These findings resonate with classic information system work on cognitive offloading and decision aids. Todd and Benbasat (1992) show that when decision support systems present highly processed information, decision makers often simplify their own strategies, potentially neglecting cues they would have considered otherwise. Similarly, explanation and transparency have been identified as key design levers to keep users cognitively engaged and to calibrate trust (Gregor and Benbasat, 1999; Komiak and Benbasat, 2006; Adomavicius et al., 2022).

Another line of work examines how digital technologies affect inequality in access, engagement, and outcomes. Early research documented that technology-mediated learning can either narrow or widen performance gaps depending on access, prior skills, and instructional design (Leidner and Jarvenpaa, 1993; Alavi and Leidner, 2001). More recent large-scale experiments on educational and training platforms show that personalized interventions, reminders, and gamification can generate

heterogeneous treatment effects across ability and motivation segments (Huang et al., 2021; Leung et al., 2023). At the same time, the algorithmic fairness literature emphasizes that algorithmic systems can create unequal benefits or harms across groups if not carefully designed and monitored (Kallus, Mao, and Zhou, 2022). Together, these perspectives highlight that AI-enabled learning tools may either mitigate or exacerbate existing disparities, depending on design, access, and user behavior.

Our study speaks directly to these cognitive and distributional concerns. Conceptually, we adopt the view that for learning, the crucial distinction is not merely *whether* AI is used, but *when* and *how*. By constraining generative AI to a post-solution debriefing role, we preserve the independent problem-solving phase prior to AI intervention while leveraging AI’s strengths in error analysis and personalized explanation afterward. Empirically, we show that this design produces steeper learning curves: conditional on total study time, students with access to post-solution AI tutoring experience larger gains in correctness over time than comparable students without access, suggesting accelerated skill acquisition rather than shallow performance gains.

Furthermore, we provide evidence on distributional effects. In contrast to concerns that AI may primarily benefit already strong students or those in high-income environments, we document that our post-solution AI tutor *closes* gaps between low- and high-performing students and does not widen disparities across income levels. In particular, low-performing students register larger improvements in both accuracy and efficiency, and proxies for local socioeconomic status do not indicate disproportionately higher gains from AI access. In this sense, our results complement prior literature showing that carefully designed technology-mediated learning can support weaker learners (e.g., Alavi, 1994; Santhanam, Sasidharan, and Webster, 2008; Leung et al., 2023) and extend it by showing that generative AI, when integrated at the debriefing stage, can promote both efficiency and equity in learning outcomes.

3 Data Description and Field Setting

3.1 Institutional Setting and Platform

Our empirical analysis is based on a collaboration with Squirrel AI, a leading Chinese education technology firm that provides digitized learning experiences to strengthen K–12 students’ mastery of the school curriculum. Founded with the mission of making high-quality education more widely

accessible, Squirrel AI has grown into one of the largest providers of AI-enabled learning solutions in China. The firm’s core business model is direct-to-consumer: students typically enroll through local learning centers, where the learning center pays an upfront device-access fee to Squirrel AI for each tablet (about \$1,500–\$2,000) and students pay an annual subscription fee (around \$1,000)¹ that is shared between headquarters and the learning centers. As of 2024, Squirrel AI reports approximately 112,000 active paying users.

Each learning center is operated by local staff and teachers, with team size varying based on enrollment. Their role is primarily administrative and supportive: they manage daily operations, advertise and recruit new students, monitor attendance and retention, and relay feedback to headquarters about the system’s performance. By contrast, the core learning technologies—including content, the digital practice environment, and the AI tutor—are centrally developed and managed by Squirrel AI headquarters. This centralization ensures that the instructional content, mastery models, and AI-enabled features are standardized across centers, while local staff focus on logistics and student support rather than instructional roles. Students interact with the learning platform primarily through a dedicated tablet-like device. The system delivers curriculum-aligned problems, records students’ intermediate work and final answers, evaluates their correctness, and then presents a detailed step-by-step solution.

3.2 Data Description

Our dataset is drawn as a random panel from the platform and covers detailed daily interaction logs from 18,904 students (who were active at the introduction of the AI tutor) across multiple centers in 2024, yielding 2,114,829 student-day observations. For each student-day, we observe (i) how many learning sessions and problems the student completed, (ii) the difficulty and correctness of each problem, (iii) time spent on problems, and (iv) whether the post-solution AI tutor mode was available and used. We also observe standard student attributes such as enrolled grade and the learning center attended.

A key feature of our setting is that AI-enabled devices were rolled out unevenly across students due to capacity constraints. Not all tablets in a center were upgraded to the AI mode, which created substantial quasi-random variation in exposure to the AI tutor both across students and over time. Some students never had access to AI-enabled devices during our sample period, while others inter-

¹Importantly, this fee allows students to gain the right to use a center-provided tablet-like device during their sessions rather than owning the device.

mittently used devices with the AI tutor activated. This resource-constrained rollout provides the variation that underlies our difference-in-differences analysis and our study of heterogeneous effects. Because all students paid the same fee for access to the learning center, centers with more students than AI-tutor tablets were instructed to allocate the AI-enabled devices randomly across students. Consistent with this guidance, our pre-treatment descriptive statistics show that students who ever used the AI-tutor mode and those who never used it are very similar in terms of the number of problems solved, average correctness rate, and average problem difficulty, although we do observe a statistically significant but economically modest difference in the average time spent per problem. That said, the platform could not micromanage or verify how each center implemented this allocation in practice, and logistical constraints may have prevented perfectly random assignment. We therefore treat the setting as quasi-random rather than a fully controlled field experiment, and we complement our difference-in-differences analysis with robustness checks, including IV regressions, to assess the sensitivity of our findings to departures from strict randomization.

Table 1 reports descriptive statistics for the main variables. Panel A summarizes student-level characteristics. The average student is in grade 6.7 (median 7). Of the 18,904 students, 12,435 used AI-tutor mode at least once during the sample period; we refer to these students as the *treated* group, and to students who never used AI-tutor mode as the *control* group. Conditional on being treated, AI-enabled devices were used for, on average, 18% of a student’s visits after the AI-tutor rollout, with substantial cross-student variation. This share reflects both demand-side factors (e.g., the number of students enrolled in a learning center who are interested in using AI-enabled devices) and supply-side constraints (e.g., the number of AI-enabled devices).

Panel B reports statistics at the student-day level. On a typical active day, students solve about 115 problems, corresponding to 1.3 hours of work. The median student-day involves 57 problems and roughly 0.7 hours of engagement. The average correctness rate is 0.74, with substantial dispersion: the inter-quartile range runs from 0.66 to 0.84. Average problem difficulty (measured as incorrectness ratio across students in the initial submissions before AI feedback) is around 61.5 on a 0–100 scale, suggesting that the platform tends to assign moderately challenging problems. Average time per problem is about 53 seconds. Taken together, these statistics highlight that the platform generates intensive practice episodes with rich variation across students and days, which we exploit to study how generative AI affects engagement, efficiency, and learning.

Figure 1 plots average outcomes for treated and control students before and after the AI-tutor rollout. Prior to rollout, the two groups exhibit similar trajectories. After rollout, however,

Table 1: Summary Statistics (Unit of Observation: Student–Date Level)

Panel A: Student level ($N = 18,904$)						
Variable	Mean	Median	Percentiles			
			5%	25%	75%	95%
Grade	6.70	7	3	5	8	10
Treated (ever AI-tutor user)	0.66	1	0	0	1	1
Pr(AI Mode On Treated=1)	0.18	0.14	0.02	0.08	0.25	0.47

Panel B: Student–day level ($N = 2,114,829$)						
Variable	Mean	Median	Percentiles			
			5%	25%	75%	95%
Problems solved	115.13	57	5	25	127	366
Avg. time per problem (seconds)	52.85	42.81	18.42	30.16	62.13	118.69
Total time (hours)	1.31	0.71	0.07	0.32	1.55	3.89
Avg. difficulty (0–100)	61.49	62.49	40.12	54.92	68.83	80.13
Correctness rate	0.74	0.76	0.45	0.66	0.84	0.94

Notes: Panel A reports student-level characteristics, including an indicator for ever using AI-tutor mode and, for treated students, the average share of visits with AI-tutor mode active after rollout. Panel B reports daily learning outcomes at the student-day level, including productivity (problems solved and time spent), problem difficulty, and correctness rates.

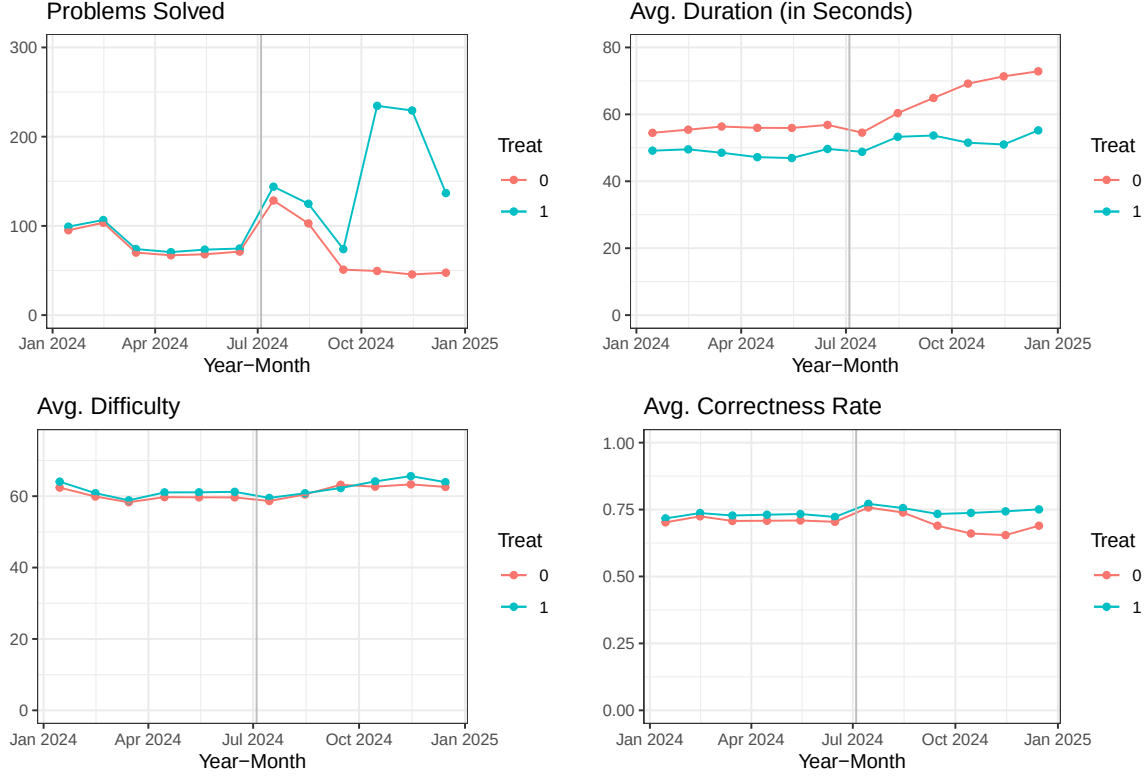
treated students show higher engagement (more problems solved per day), shorter average time per problem, and notably higher correctness rates. In contrast, we do not observe pronounced divergence in average difficulty, suggesting that the platform maintains comparable challenge levels across groups while AI reshapes how effectively students convert practice into correct answers.

3.3 Digital Experience: Baseline vs. Post-Solution AI Tutor

To interpret our empirical results, it is useful to compare the baseline digital practice environment to the AI-tutor mode.

In the *baseline mode* (no AI tutor), students work on curriculum-aligned math problems delivered through the tablet interface. A typical example, shown in Figure 2(a), is: “One type of wire costs 12.5 yuan per meter. Uncle Li bought 2.7 meters. How much does it cost in total? (Retain one decimal place).” As illustrated in Figure 2(b), students see a numerical fill-in-the-blank question, enter an answer, and submit it. The system immediately reports whether the answer is correct or incorrect and then displays a complete, step-by-step solution. Students can review this solution and then proceed to the next question. This design resembles a traditional digital homework or quiz environment: the interaction is limited to correctness feedback and a static worked-out solution,

Figure 1: Model-Free Evidence on the Effects of AI-Tutor Mode

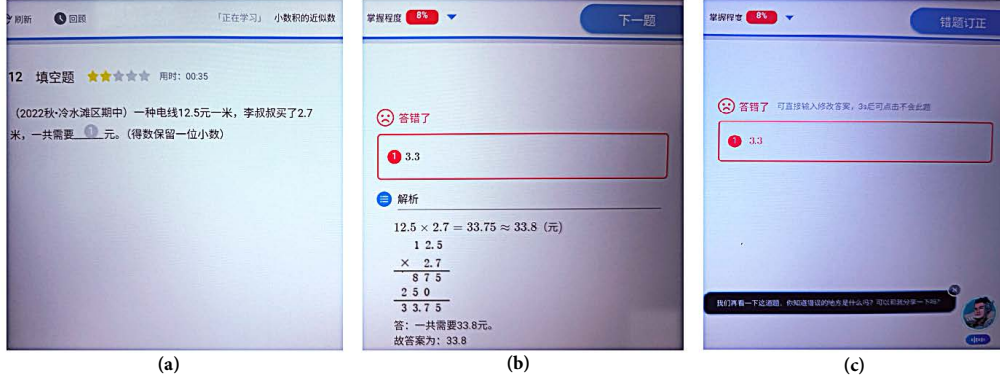


without personalized dialogue or adaptive follow-up.

The *AI-tutor mode* enriches this workflow by adding a post-solution generative AI tutor that activates after a student submits an answer. Importantly, the problem presentation itself, shown in Figure 2(a), is identical to the baseline mode. Similarly, as depicted in Figure 2(c), the answer-entry interface is nearly identical. However, the key difference for AI users is an integrated AI chat panel. Once an answer is submitted, the AI opens a debrief panel that (i) reports correctness, (ii) highlights which steps or concepts appear to be problematic (if any), and (iii) engages the student in an interactive dialogue. The AI tutor leverages the student’s historical performance and current attempts—including digital “scratches” or drafts—to infer the likely sources of error and tailor explanations accordingly. Rather than simply displaying a fixed solution, the AI tutor offers targeted hints, step-by-step explanations, and real-time responses to follow-up questions, providing a more personalized and adaptive learning experience.

Figure 3 provides a more detailed illustration of the AI tutor workflow. The student first attempts the problem and submits an initial answer (panel (a)). Upon submission, the AI tutor

Figure 2: Comparison of Baseline and AI-Tutor Modes



Notes: Panel (a) illustrates how students—with and without AI-tutor access—approach a given problem. In the baseline mode (panel (b)), students submit an answer, receive correctness feedback, and then see the correct solution. With AI-tutor access (panel (c)), submitting an answer triggers an AI debrief that diagnoses errors, answers follow-up questions, and provides the correct solution with step-by-step explanations.

automatically opens a debrief panel that confirms correctness, offers targeted guidance, and responds to clarifying questions (panels (b) and (c)). The student can then revise the approach and resubmit; the tutor iteratively clarifies the original mistake, walks through the key steps of the correct solution, explains the underlying rationale, and addresses remaining questions (panel (d)). Crucially, AI assistance is strictly *post-solution*: students must commit to an answer before the AI intervenes. This post-solution design preserves independent problem solving while enabling generative AI to transform mistakes into structured learning opportunities.

Figure 3: Detailed Workflow for a Student with AI-Tutor Access



Notes: This figure presents a more detailed, step-by-step illustration of the AI-tutor workflow. Panel (a) shows the student's initial attempt and submission; panels (b) and (c) display the AI tutor's debrief with feedback and interaction; panel (d) shows the iterative clarification and explanation process.

4 Main Effects of AI Tutor Mode

In this section, we describe our empirical models and identifying assumptions for the main analysis and then present the corresponding results.

4.1 Estimation and Identification Strategy

Let i, t denote the individual and date. Our main specification for the main effects of the AI tutor on student outcomes is as follows:

$$Y_{i,t} = \beta (Treat_i \cdot Post_t) + \alpha_i + \delta_t + e_{i,t}, \quad (1)$$

where $Y_{i,t}$ denotes the outcome variables of interest, such as the number of problems solved, average duration, and average correctness ratio, for individual i on date t . As is standard in a difference-in-differences analysis, $Treat_i$ is an indicator for whether student i ever received treatment, and $Post_t$ is an indicator for the period after the introduction of the AI tutor mode. α_i and δ_t represent student and date fixed effects, which we further discuss below. We cluster standard errors at the student level.

Our key identifying assumption is the parallel trends assumption: without the introduction of the AI tutor mode, treated and control students would have followed similar trends in the outcome variables of interest, $Y_{i,t}$. The model-free patterns in Figure 1 seem to support this assumption. We further strengthen identification by controlling for time-invariant individual-level heterogeneity through α_i (e.g., baseline ability or work ethic) and time-varying common shocks through δ_t (e.g., semester timing). Under the parallel trends assumption, β identifies the causal effect of the AI tutor mode on learning outcomes.

Additionally, to assess the plausibility of our identifying assumption and to estimate time-varying effects, we estimate:

$$Y_{i,t} = \sum_{k \neq -1} \beta_k (Treat_i \cdot \mathbf{1}\{t - t_0 = k\}) + \alpha_i + \delta_t + e_{i,t}, \quad (2)$$

where $\mathbf{1}\{t - t_0 = k\}$ is an indicator equal to 1 if date t is k months away from the month when the AI tutor mode was introduced (t_0). We omit $k = -1$ (the last pre-treatment month) as the reference month.

Each β_k measures the treated-control difference in $Y_{i,t}$ at relative month k , relative to the omitted month, while controlling for student fixed effects (α_i) and date fixed effects (δ_t). Under

Table 2: Main Effects of AI Tutor on Learning Outcomes (GRADUALLY)

	Problems solved (log) (1)	Avg. time (seconds) (2)	Avg. correctness rate (3)
Treat · Post	0.3057*** (0.0103)	-2.1779*** (0.4024)	0.0264*** (0.0011)
Avg. Difficulty	0.0122*** (0.0001)	-0.2497*** (0.0054)	0.0048*** (0.00002)
Student FE	Y	Y	Y
Date FE	Y	Y	Y
Adjusted R^2	0.2711	0.2187	0.2821
N	2,114,829	2,114,829	2,114,829

Notes: S.E.s clustered at the student level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

the parallel trends assumption, pre-treatment coefficients ($k < 0$) are expected to be around zero, while post-treatment coefficients ($k \geq 0$) capture the dynamic response following the AI tutor introduction.

4.2 Results

Table 2 reports the estimates from Equation (1). Each column corresponds to a difference-in-differences regression with student and date fixed effects, where the key regressor is the interaction Treat · Post. First, column (1) in Table 2 shows a large engagement response. The coefficient 0.3057 on Treat · Post implies that treated students increase the number of problems solved per day by approximately 35.8% relative to similar control students ($\exp(0.3057) - 1 \approx 0.358$). Column (2) indicates that treated students spend less time per problem after the AI rollout: the coefficient -2.1779 implies a reduction of about 2.18 seconds per problem. Given that the average post-period time per problem for controls is about 55 seconds, this corresponds to roughly a 3.9% decline, suggesting efficiency gains rather than mere reallocation of effort. Next, column (3) shows that performance accuracy (i.e., correctness ratio) also improves. The coefficient 0.0264 indicates that the average daily correctness rate increases by 2.64 percentage points for treated students relative to controls. Given that baseline post-treatment correctness is around 74%, this represents about a 3.6% improvement. Taken together, these results show that access to the post-solution generative AI tutor leads students to practice more, solve problems faster, and answer a higher fraction of them correctly.

Figure 4 presents the estimates corresponding to Equation (2). The pre-treatment coefficients

for the number of problems solved (panel (a)) are close to zero, supporting the parallel-trends assumption. Following the introduction of the AI tutor, treated students’ problem volume rises steadily, peaking about four months after rollout and then stabilizing at a higher level. For average time per problem and correctness (panels (b) and (c)), we observe small but noticeable deviations from flat pre-trends, which we interpret cautiously. Post-treatment, however, the patterns are clear: treated students’ time per problem declines and correctness increases, with the effects growing over the first few months and stabilizing thereafter. At their peak, treated students spend roughly 10 seconds less per problem and achieve correctness rates about 5–6 percentage points higher than controls. Overall, the dynamic analysis shows that the benefits of AI tutoring ramp up over time as students learn how to leverage the tool, and as the system accumulates data to personalize feedback.

Although the main effects indicate positive outcomes on average, they may mask substantial heterogeneity. We next examine which students benefit the most and whether AI tutoring helps close or widen existing educational gaps, by focusing on heterogeneity by prior performance and by income.

5 Heterogeneity Effects and Potential Mechanisms

In this section, we examine the heterogeneous effects by prior performance (before the introduction of AI) and by income, proxied by local disposable income. We then explore the potential mechanisms underlying these effects.

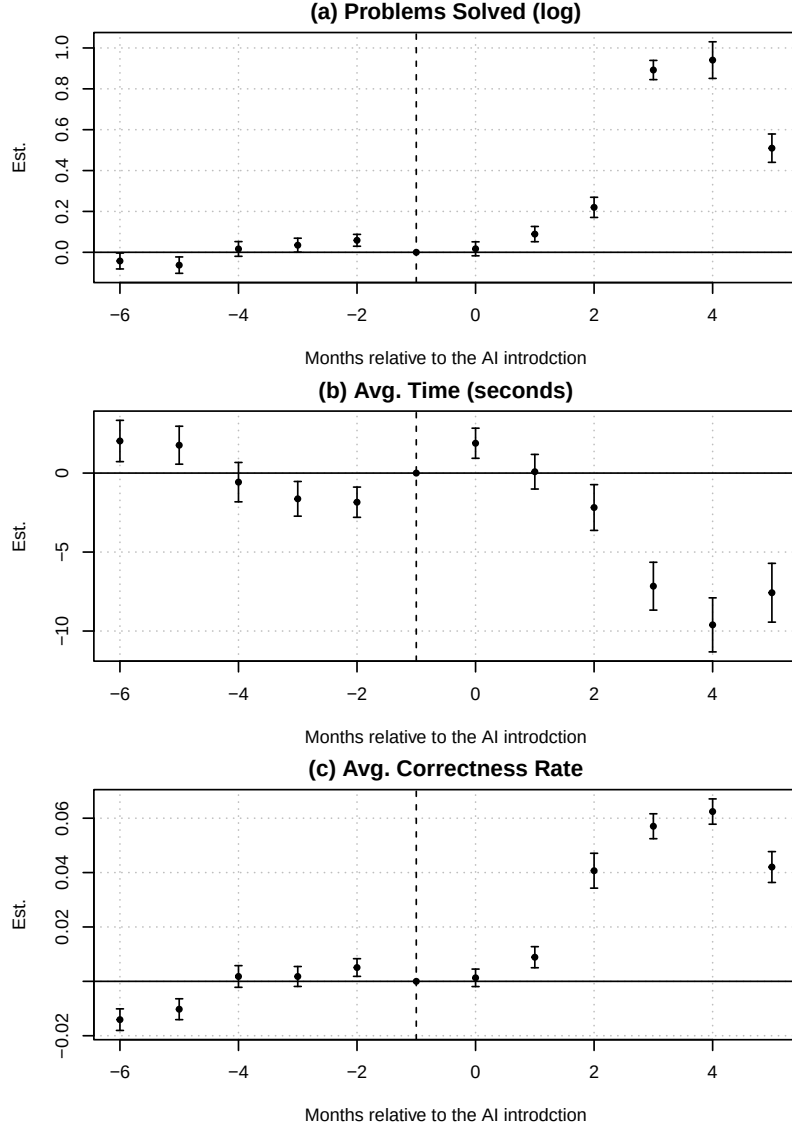
5.1 Heterogeneity Effects

Motivated by the possibility that AI may either promote or exacerbate equity in education, we examine how effects vary by prior performance and by county-level income. Specifically, we estimate a triple difference specification:

$$Y_{i,t} = \beta_0 (\text{Treat}_i \cdot \text{Post}_t) + \sum_{g \neq g_0} \beta_g (\text{Treat}_i \cdot \text{Post}_t \cdot \mathbf{1}\{i \in g\}) + \alpha_i + \delta_t + \sum_{g \neq g_0} \gamma_g (\text{Post}_t \cdot \mathbf{1}\{i \in g\}) + e_{i,t}, \quad (3)$$

where $\mathbf{1}\{i \in g\}$ is an indicator for the membership of student i in group $g \in \mathcal{G}$ (e.g., income or performance groups), and g_0 is the reference group. We include student fixed effects (α_i) and date

Figure 4: Dynamic Effects on Learning Outcomes



Notes: Each dot and bar represent the point estimate and the 95% confidence interval.

fixed effects (δ_t). Additionally, the terms γ_g allow groups to have different post-period level shifts (common to treated and control students within each group).

The coefficient β_0 captures the average effect of the AI tutor for the reference group g_0 . For any $g \neq g_0$, β_g is a triple-difference term that measures how the treatment effect for group g differs from the effect for g_0 . Standard errors are clustered at the student level.

5.1.1 Heterogeneity by prior performance

We first split students according to their pre-treatment correctness ratio. Let $\mathbf{1}\{\text{Low}\}$ indicate that student i 's pre-treatment correctness is in the bottom third of the distribution, and $\mathbf{1}\{\text{Mid}\}$ indicate the middle third (with high performers as the omitted group).² Table 3 reports the results, corresponding to Equation (3). Note that the baseline $\text{Treat} \cdot \text{Post}$ coefficient captures the effect for high-performing students.

Column (1) in Table 3 shows that even high performers increase their problem volume by about 28.6% (in log terms) after gaining access to the AI tutor. The interaction term for low performers is positive and statistically significant: low-performing students experience an additional 0.0739 log-point increase relative to high performers. Combining the baseline and interaction, the total effect for low performers is roughly $0.2864 + 0.0739 \approx 0.360$, or about a 43% increase in practice volume. Thus, low-performing students respond even more strongly in engagement than high performers.

Column (2) reveals even starker heterogeneity in correctness. High performers increase their correctness by about 2.4 percentage points, while low performers gain an additional 1.65 percentage points, for a total of roughly 4.0 percentage points. Given that low performers begin from much lower baseline accuracy, this is a sizable relative improvement, indicating that the AI tutor helps them partially close the existing performance gap. Mid-performing students' gains are similar to those of high performers. Overall, AI tutoring appears especially beneficial for those who need help the most, consistent with a "catch-up" or equalizing effect.

5.1.2 Heterogeneity by income

We next study whether AI tutoring equalizes or exacerbates outcome differences by income. Specifically, we proxy each student's economic status using county-level disposable income and construct terciles: low, middle, and high income.

Table 4 reports estimates where $\mathbf{1}\{\text{Low}\}$ and $\mathbf{1}\{\text{Mid}\}$ indicate students from low- and middle-income counties, respectively (high income is the omitted group).³ Column (1) shows that the low- and middle-income interactions are close to zero and statistically insignificant, implying that students from these areas increase their problem-solving activity by roughly the same proportion as those from high-income areas. For correctness, presented in column (2), the middle-income in-

²We measure pre-treatment performance using the average correctness ratio across days when a student attended the learning center. Weighting by the number of problems yields very similar results.

³We note that the sample size changes in this analysis, as for some students we are unable to obtain county-level income data from the local statistics bureau.

Table 3: Heterogeneous Effects by Past Performance

	Problems solved (log) (1)	Avg. correctness rate (2)
Treat · Post	0.2864*** (0.0183)	0.0238*** (0.0017)
Treat · Post · 1{Low}	0.0739*** (0.0256)	0.0165*** (0.0027)
Treat · Post · 1{Mid}	0.0251 (0.0252)	0.0021 (0.0024)
Avg. Difficulty	0.0120*** (0.0001)	0.0048*** (0.00002)
Individual FE	Y	Y
Date FE	Y	Y
Post × Perf. Group Controlled	Y	Y
Adjusted R^2	0.2720	0.2852
N	2,114,829	2,114,829

Notes: S.E.s clustered at the student level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

teraction is positive and small (0.006), while the low-income interaction is essentially zero. Overall, all income groups experience similar accuracy gains. Hence, AI tutoring does not appear to disproportionately favor already advantaged students; if anything, it provides comparable benefits across the income distribution, which is encouraging from an equity standpoint.

5.2 Potential Mechanisms

Next, we examine potential mechanisms through which AI-assisted learning affects outcomes. On one hand, exposure to the AI tutor may make study more engaging and enjoyable, thereby increasing the likelihood that students attend the learning center (i.e., extensive-margin effect). On the other hand, conditional on attendance, access to AI may help students learn more effectively and efficiently (i.e., intensive-margin effect).

For the attendance rate (extensive margin), we estimate the following model:

$$Y_{i,t} = \beta (Treat_i \cdot Post_t) + \alpha_i + \delta_t + e_{i,t}, \quad (4)$$

where the outcome variable, $Y_{i,t}$, is (a) an indicator for whether student i attended at least once in week t , or (b) the number of attendances in week t , either starting from our sample period or since student i joined, whichever comes first. As above, α_i denote student fixed effects and δ_t denote week fixed effects. We cluster standard errors at the student level.

Table 4: Heterogeneous Effects by Income

	Problems solved (log) (1)	Avg. correctness rate (2)
Treat · Post	0.312*** (0.008)	0.025*** (0.001)
Treat · Post · 1{Low}	0.008 (0.012)	0.001 (0.001)
Treat · Post · 1{Mid}	-0.004 (0.012)	0.006*** (0.001)
Avg. Difficulty	0.012*** (0.0001)	0.005*** (0.00001)
Student FE	Y	Y
Date FE	Y	Y
Post × Inc. Group Controlled	Y	Y
Adjusted R^2	1,390,545	1,390,545
N	0.277	0.285

Notes: S.E.s clustered at the student level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To study within-student changes in learning on days when students attend the center (intensive margin), we estimate:

$$Y_{i,t} = \beta AI_{i,t} + \alpha_i + \delta_t + e_{i,t}, \quad (5)$$

where $Y_{i,t}$ denotes daily outcomes (e.g., number of problems solved, correctness ratio), and $AI_{i,t}$ is an indicator that student i used the AI-assisted tablet on date t . α_i and δ_t denote student fixed effects and date fixed effects, respectively. Standard errors are clustered at the student level.

Because students may strategically choose to learn with AI on some days when they feel unusually productive or motivated (e.g., before an upcoming exam), the naive OLS estimate $\hat{\beta}$ in (5) may be biased. We therefore instrument for $AI_{i,t}$ using capacity-driven AI availability at the student's center $c(i)$ on date t . In particular, many centers had a limited number of AI-enabled devices relative to typical attendance. Conditional on student and date fixed effects, capacity-/congestion-driven fluctuations in the availability of AI tablets induce quasi-random variation in access to AI.

Specifically, we construct two instruments using only information from the prior 15 days (to avoid simultaneity), excluding date t . Let $\mathcal{W}_{15}(t) = \{t - 15, \dots, t - 1\}$. Define:

$$Z_{c(i),t}^{(1)} = \max_{\tau \in \mathcal{W}_{15}(t)} \{\text{AI devices}_{c(i),\tau}\}, \quad Z_{c(i),t}^{(2)} = \frac{\max_{\tau \in \mathcal{W}_{15}(t)} \{\text{AI devices}_{c(i),\tau}\}}{\max_{\tau \in \mathcal{W}_{15}(t)} \{\text{Total devices}_{c(i),\tau}\}}.$$

For the validity of the proposed instruments, it is high plausible that greater AI capacity at a

Table 5: Effects of Using AI on the Extensive Margin (Attendance)

	Whether Attended (1)	Number of Attendance (2)
Treat · Post	0.0403*** (0.0045)	0.1424*** (0.0242)
Student FE	Yes	Yes
Week FE	Yes	Yes
Adjusted R^2	0.2681	0.3984
N	764,020	764,020

Notes: S.E.s are clustered at the individual level; $*p < 0.1$; $**p < 0.05$; $***p < 0.01$.

center increases take-up (relevance), while, conditional on fixed effects (α_i and δ_t), lagged center-level capacity affects $Y_{i,t}$ only through AI use, $AI_{i,t}$ (exclusion).

5.2.1 Extensive Margin (Attendance)

Table 5 shows the results corresponding to Equation (4), where we aggregate the data to the weekly level to explore weekly attendance outcomes. Column (1) considers a binary indicator for whether a student attended at least once in a given week; column (2) uses the number of days attended. The probability of attending at least once in a week increases by 0.0403 for treated students in the post period. Relative to the baseline weekly attendance rate of approximately 0.76, this corresponds to a 5.3% increase. The number of attendance days per week rises by 0.1424 days, which is about a 5.1% increase relative to the mean of 2.77 days.

5.2.2 Intensive Margin (Within-Session Productivity)

We next study what happens conditional on attending. Table 6 reports OLS and IV estimates, corresponding to Equation (5), where the key regressor is an indicator for whether the student used the AI tutor on a given day. The outcome variables are the number of problems solved (in log) and the correctness rate. The OLS estimates (columns (1) and (2)) suggest very large differences: on days when students use the AI tutor, they solve $\exp(1.903) - 1 \approx 5.71$ times more problems (about a 570% increase) and have correctness about 6.1 percentage points higher than on non-AI days. These estimates are likely upwardly biased due to selection (e.g., students may be more motivated or working on different types of content on AI days).

The IV estimates, which use center-level capacity-driven availability as an instrument (either

Table 6: Heterogeneous Effects of AI on Learning Outcomes

	OLS		IV			
	Problems (in log) (1)	Correctness (2)	Problems (in log) (3)	Correctness (4)	Problems (in log) (5)	Correctness (6)
AI on	1.903*** (0.006)	0.061*** (0.001)	1.394*** (0.078)	0.015* (0.008)	1.563*** (0.098)	0.022** (0.011)
Avg. Difficulty	0.014*** (0.0002)	0.004*** (0.00003)	0.012*** (0.0003)	0.004*** (.00003)	0.013*** (0.0004)	0.004*** (0.00005)
F-Statistics			9,767.9	9,767.9	4,762.2	4,762.2
Student FE	Y	Y	Y	Y	Y	Y
Date FE	Y	Y	Y	Y	Y	Y
Adjusted R^2	0.434	0.272	0.421	0.264	0.429	0.267
N	948,825	948,825	948,825	948,825	948,825	948,825

Notes: S.E.s are clustered at the individual level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

the number of AI-enabled devices or the ratio of AI-enabled devices to total devices over the past 15 days), provide more conservative estimates. In column (3), the coefficient 1.394 implies that using the AI tutor on a given day increases the number of problems solved by about 303% ($\exp(1.394) - 1 \approx 3.03$), meaning it roughly quadruples daily problem-solving. In column (4), correctness increases by about 1.5 percentage points. Columns (5)–(6), which use the ratio instrument, yield similar qualitative conclusions. In short, conditional on attending, AI use dramatically increases within-session productivity and modestly improves accuracy.

6 AI Tutor Usage and Learning Curves

In this section, we further explore the returns to AI on education outcomes. First, we explore whether the returns to AI-assisted learning today depends on a student’s level of recent AI usage in a relatively short-run. Next, we examine how the AI tutor affects longer-term skill development.

6.1 Diminishing Returns from Over-Reliance on AI

To investigate whether the effect of AI use depends on a student’s recent history of AI exposure, we allow the impact of using AI on a given day to vary with cumulative prior use. This specification enables us to test for diminishing returns or complementary learning dynamics in how students engage with AI over time. Accordingly, we estimate:

$$Y_{i,t} = \beta AI_{i,t} + \phi CumAI_{i,t-1} + \theta (AI_{i,t} \cdot CumAI_{i,t-1}) + \alpha_i + \delta_t + e_{i,t}, \quad (6)$$

where $AI_{i,t}$ is an indicator that student i used AI on date t , and $CumAI_{i,t-1}$ is the cumulative number of days with AI use in the previous 2 weeks (excluding t). Standard errors are clustered at the student level.

Note that $\theta < 0$ indicates diminishing returns to AI with higher prior usage (i.e., the benefit of using AI today is reduced as past AI use accumulates), whereas $\theta > 0$ indicates complementarity or learning (i.e., the benefit increases with prior usage).

Table 7 reports the results, corresponding to Equation (6). The interaction term AI on·Cum. AI is negative and statistically significant in both the OLS and IV specifications for problems solved and correctness, where we use center-level capacity-driven availability as an instrument (either the number of AI-enabled devices or the ratio of AI-enabled devices, as before). Focusing on the IV estimates (columns (3)–(6)), the coefficient -0.283 in column (3) implies that the log effect of using AI on a given day declines with recent AI use. For example, if a student has used AI on 10 of the prior 14 days, the incremental log increase in problems solved from using AI today is reduced by about $0.283 \times 10 = 2.83$ relative to a student with no recent AI use. Similarly, the coefficient -0.056 in column (4) indicates that the accuracy gain from AI use shrinks as cumulative recent usage rises and can become negligible or even slightly negative at very high levels of past use. Finally, these estimates are very similar when we use the alternative IV based on the ratio of AI-enabled devices (columns (5)–(6)).

Overall, these results suggest diminishing marginal returns to AI usage in the short run: students who rely heavily on AI every day see smaller incremental benefits from using it again today than students who use it more intermittently. Overall effects remain positive on average, but the marginal gain per additional AI session is lower when the student has just used AI very frequently. This points to the importance of balanced usage and raises the possibility that mixing AI-supported practice with unaided practice may be more effective than continuous, high-frequency AI assistance.

6.2 The Impact of AI on Longer-Term Skill Development

Finally, we explore whether the AI tutor improves not only contemporaneous outcomes but also the rate at which students learn over time. To do so, we estimate a following specification:

Table 7: Effects of Past AI Usage

	OLS		IV			
	Problems (in log)	Correctness	Problems (in log)	Correctness	Problems (in log)	Correctness
	(1)	(2)	(3)	(4)	(5)	(6)
AI on	1.904*** (0.009)	0.077*** (0.001)	1.400*** (0.122)	0.014 (0.016)	1.482*** (0.169)	-0.008 (0.029)
Cum. AI	0.001 (0.001)	0.004*** (0.0001)	0.184*** (0.042)	0.037*** (0.006)	0.176** (0.070)	0.043*** (0.013)
AI on · Cum. AI	-0.0004 (0.001)	-0.004*** (0.0002)	-0.283*** (0.072)	-0.056*** (0.011)	-0.275** (0.119)	-0.066*** (0.023)
Avg. Difficulty	0.014*** (0.0002)	0.004*** (0.00003)	0.006*** (0.002)	0.003*** (0.0002)	0.007*** (0.003)	0.003*** (0.0005)
F-Statistics			2,839.1	2,839.1	1,286.1	1,286.1
Student FEs	Y	Y	Y	Y	Y	Y
Date FEs	Y	Y	Y	Y	Y	Y
Adjusted R^2	0.434	0.275	0.186	-0.278	0.213	-0.546
N	955,385	955,385	948,825	948,825	948,825	948,825

Notes: S.E.s are clustered at the individual level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

$$\begin{aligned}
\text{correctness}_{it} = & \beta_1 \text{cumul. hours}_{it} \\
& + \beta_2 (\text{cumul. hours}_{it} \cdot \text{Post}_{it}) + \beta_3 (\text{cumul. hours}_{it} \cdot \text{Treatment}_i) \\
& + \beta_4 (\text{cumul. hours}_{it} \cdot \text{Treatment}_i \cdot \text{Post}_{it}) \\
& + \beta_5 \text{pre-treat cumul. hours}_{it} + \beta_6 (\text{Treatment}_i \cdot \text{Post}_{it}) + \beta_7 \text{avg difficulty}_{it} \\
& + \alpha_i + \lambda_t + \varepsilon_{it},
\end{aligned} \tag{7}$$

where correctness_{it} is the student's daily correctness rate (multiplied by 100). cumul.hours_{it} denotes the cumulative number of hours student i has studied up to date t if t is before the introduction of AI (i.e., $\text{Post}_{it} = 0$), or the cumulative number of hours student i has studied *since* the AI rollout (i.e., $\text{Post}_{it} = 1$), which resets to zero at the moment of AI introduction. $\text{pre-treat cumul. hours}_{it}$ equals zero during the pre-treatment period ($\text{Post}_{it} = 0$), and equals the total number of hours student i accumulated during the pre-treatment period when $\text{Post}_{it} = 1$. This is to capture a structural break in the cumulative measure. The remaining variables are defined as in the earlier specifications. Standard errors are clustered at the student level.

The key coefficient of interest is β_4 on the triple interaction term ($\text{cumul. hours} \cdot \text{Treat} \cdot \text{Post}$),

which captures whether treated students convert additional study hours into correctness gains more efficiently than control students after AI rollout.

Table 8 presents the results for the full sample and for terciles of pre-treatment cumulative hours (Low, Mid, High). In the full sample (column (1)), the triple interaction is positive and highly significant (0.027). Since the outcome is correctness rate $\times 100$, this indicates that, after the AI rollout, each additional hour of study yields an extra 0.027 percentage point improvement in correctness for treated students relative to controls. Over many hours, these incremental gains accumulate. For example, over 50 hours of post-treatment study, treated students’ correctness would improve by roughly 1.35 percentage points more than that of comparable control students. This provides direct evidence that AI tutoring steepens learning curves: students with AI convert practice into mastery more efficiently.

Columns (2)–(4) show that this positive effect on learning rates is present across all terciles of pre-treatment hours. The triple interaction is positive and statistically significant for low-, mid-, and high-hour groups, with magnitudes in a fairly similar range (0.017–0.029). Thus, AI tutoring enhances learning efficiency not only for students who had already accumulated substantial experience on the platform but also for those with relatively little pre-treatment exposure.

Importantly, these learning-curve results complement the within-period findings. They show that AI tutoring does more than simply boost contemporaneous performance or allow students to “game” the system. Instead, restricting AI assistance to post-solution debriefs—after students attempt the problem on their own—appears to promote deeper learning: students learn more from each hour of practice, and these gains cumulate into superior performance over time.

7 Discussion

Our investigation provides encouraging evidence that AI tools can be integrated into education in a way that boosts engagement and learning effectiveness without sacrificing the long-term development of skills. This stands in contrast to more cautionary findings from settings where AI is available during problem-solving and effectively performs too much of the cognitive work for students. The key distinction in our context is *when* and *how* the AI is applied: by acting as a tutor *after* the student’s attempt (rather than a shortcut *during* the attempt), the AI supports learning rather than short-circuiting it. In combination with the results in Section ??, this suggests that carefully designed “post-solution” AI tutoring can increase engagement, efficiency, and accuracy

Table 8: Effects of AI on learning

	Correctness rate $\times 100$			
	Full (1)	Low (2)	Mid (3)	High (4)
Cumul. hours	-0.039*** (0.002)	-0.040*** (0.005)	-0.037*** (0.004)	-0.028*** (0.003)
Cumul. hours $\cdot$ Post	0.004 (0.003)	0.014** (0.006)	0.002 (0.005)	-0.007 (0.005)
Cumul. hours $\cdot$ Treatment	0.011*** (0.002)	0.016*** (0.004)	0.011*** (0.003)	0.006** (0.003)
Cumul. hours $\cdot$ Post $\cdot$ Treat	0.027*** (0.003)	0.017*** (0.006)	0.028*** (0.004)	0.029*** (0.004)
Pre-Treatment cumul. hours	-0.032*** (0.002)	-0.047*** (0.004)	-0.027*** (0.002)	-0.009*** (0.003)
Avg. Difficulty	0.482*** (0.002)	0.527*** (0.005)	0.495*** (0.003)	0.394*** (0.004)
Individual FEs	Y	Y	Y	Y
Date FEs	Y	Y	Y	Y
Post $\times$ Treat Group Controlled	Y	Y	Y	Y
Observations	2,114,829	614,132	810,134	690,563
Adjusted R^2	0.284	0.271	0.200	0.196

Notes: S.E.s are clustered at the individual level; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

while also improving longer-term skill development.

7.1 Implications for the Human–AI Interaction in Education

One of the broadest implications of our study concerns the ongoing debate on AI as a complement versus a substitute in skilled work. Education is a domain where independent learning has traditionally been considered essential. A common concern with AI tutors is that they might replace independent thinking and erode the role of human-driven learning. Our results instead illustrate a model of human–AI complementarity. In our setting, the AI tutor automates tasks that do not entirely substitute the independent thinking of students by providing immediate feedback, fine-grained error diagnosis, and explanations for routine problems.

7.2 Educational Inequality and AI as an Equalizer

Our heterogeneity analyses carry hopeful news regarding educational inequality: the largest gains from the AI tutor accrued to students who were initially struggling (lower prior performers). These are precisely the students most at risk of falling behind in traditional settings. The fact that they

improved more in terms of correctness and learning rates indicates that AI tutoring helped them catch up relative to higher-performing peers. From an equity perspective, these mechanisms suggest that AI tutors, if made broadly accessible, could narrow achievement gaps. However, accessibility is crucial. The optimistic scenario we document—AI narrowing gaps within our sample—implicitly presupposes that students have relatively equal access to the AI tutor once it is deployed. If AI resources are concentrated in wealthier schools or among families that can afford them, the macro-level effect could instead be gap-widening. This points to a clear policy implication: there is a strong case for investing in AI tutoring infrastructure in under-resourced schools and communities. Much like internet access or textbooks, AI tutors may become part of the basic educational toolkit that should be provided to all students to help level the playing field. Given recent estimates that high-quality AI tutoring can be delivered at modest annual per-student costs, such investments could yield substantial returns in learning.

7.3 AI During vs. After Problem-Solving

A central contribution of our study is to highlight *when* in the learning process AI intervention is most beneficial. Prior mixed results in the literature can be partly reconciled by focusing on this timing dimension. When AI is available during problem-solving (for example, when a student can ask the AI for hints or solutions at any time), there is a strong temptation to offload the difficult parts of the task to the AI at the first sign of struggle. This can lead to shallow engagement: students may quickly rely on the AI rather than working through the problem themselves. Recent evidence, such as [Bastani et al. \(2025\)](#), shows that providing generative AI during problem-solving can reduce subsequent learning, precisely because students bypass the core cognitive work. By contrast, in our setting, students could not invoke the AI tutor until after they had submitted an attempt. This seemingly small design choice has large implications. It imposes a period of independent struggle, which is central to durable learning. Only after committing to an answer—right or wrong—does the AI step in to guide the student through a debrief of their solution, identify errors, and clarify reasoning. Our empirical results show that this design can both improve immediate performance and longer-term skill development.

From a design and policy standpoint, these insights suggest that educational AI tools should incorporate deliberate friction to ensure they complement rather than substitute student effort. For example, AI homework assistants might be configured to provide detailed feedback only after the

student has completed and submitted the problem set, rather than offering step-by-step solutions in real time. When mid-problem help is allowed, AI systems could prioritize hints that nudge students toward productive strategies rather than revealing full solutions. Emerging platforms that implement such “guardrails” aim to preserve the benefits of challenge while still providing support. Our results strongly support this approach: these guardrails are not merely precautionary but appear central to achieving positive learning outcomes.

7.4 Diminishing Returns and Sustainable AI Usage

Our analysis of cumulative AI usage reveals an important nuance: although using the AI tutor is beneficial on average, there are clear diminishing returns when it is used very intensively over a short period. As students rely on AI on more days in the recent past, the incremental gains from using it again today shrink and, for accuracy, can become negligible at very high levels of recent use. This suggests that an “always-on” AI tutor is less effective at the margin than one used in a more balanced way. In practice, this points to the value of more sustainable usage patterns in which students alternate between AI-supported sessions and independent practice, treating AI feedback as a scarce, high-value resource rather than a default crutch. Platforms can facilitate this by, for example, gently limiting the number of explanations per session, requiring a minimum amount of unaided work before unlocking detailed AI debriefs, or prompting students who have used a lot of AI help to attempt the next few problems on their own. The analogy to effective classroom teaching is instructive: good teachers do not immediately provide answers, but instead allow students to struggle productively before stepping in with targeted explanations. Our evidence on diminishing returns similarly underscores that more AI is not always better; carefully calibrated assistance is likely to yield the greatest learning benefits.

7.5 Limitations and External Validity

While our findings are promising, several limitations should be acknowledged when interpreting their generalizability. First, our empirical setting is specific: private after-school learning centers in China focused on mathematics. Students and families who enroll in such centers may be more motivated or have different expectations than those in typical public-school classrooms. Classroom management, curricular constraints, and heterogeneity in baseline motivation could all affect the impact of AI tutors in other environments. Nonetheless, we believe that the core mechanisms we

identify—requiring an initial attempt, providing immediate but post-solution feedback, and personalizing debriefs—are likely to be relevant in broader contexts.

Second, our empirical design relies on a quasi-experimental rollout across centers, along with the parallel trends assumption. We have shown that pre-trends are largely parallel and that observables are balanced, and we employ rich fixed effects and robustness checks. Nonetheless, the design is not a pure randomized controlled trial, and residual unobserved differences across centers or cohorts could remain. Future work could complement our evidence with randomized trials in other settings, such as randomizing access to the AI tutor at the classroom or school level within a district.

Finally, the AI tutor we study embodies a particular design: post-solution, LLM-based debriefing within a structured practice environment. Other AI tutors may differ in their interaction patterns, content, and safeguards. Our findings therefore speak most directly to a regime of *post-solution AI feedback* rather than to all possible AI configurations. Designs that, for example, offering limited mid-problem hints might achieve similar benefits if they preserve the core principle of not allowing AI to do the central cognitive work for students. The broader takeaway is not that AI assistance during problem-solving must always be avoided, but that it must be carefully constrained so as not to “spoil” the problem. Continued experimentation with alternative designs, and rigorous evaluation across diverse educational contexts, will be essential to refining best practices for AI in education.

8 Conclusion

This study provides large-scale empirical evidence that an AI tutoring system, when used as a post-problem feedback tool, can substantially improve both the efficiency and effectiveness of student learning. Students with access to the AI tutor in our natural experiment completed far more practice, did so in less time per problem, and achieved higher accuracy rates. These immediate gains translated into faster learning over time: the AI-assisted students showed steeper improvement in performance for each hour of study, indicating genuine skill development rather than superficial performance boosts. Importantly, the benefits were most pronounced for those who typically struggle (i.e., lower-performing students), suggesting that AI tutors can function as an equalizer in education by helping those at the bottom of the achievement ladder climb up.

Our findings address a central question in the age of AI: when does AI enhance human learning, and when might it undermine it? The evidence here supports a nuanced view. AI can greatly

enhance learning when deployed in a complementary role that reinforces (rather than replaces) student effort. In our context, by requiring students to attempt problems on their own and then using AI to review and learn from mistakes, the technology augmented students’ ability to learn from practice. This stands in contrast to scenarios where AI provides answers or excessive help during problem-solving, which can undercut the learning process. Thus, the timing and design of AI assistance are critical. Education stakeholders should take note: simply adding AI to classrooms is not a guarantee of success – how it is integrated will determine outcomes.

However, our results also counsel caution against overusing AI. The diminishing returns we observed suggest it’s possible to have too much of a good thing. Educational programs should consider structuring AI usage – for instance, perhaps using AI tutors a few times a week rather than every day, or alternating between AI-guided and traditional learning to maximize benefits. Continued research and innovation can refine these guidelines. In the broader societal debate, our study provides a hopeful counterpoint to fears that AI will deskill individuals or exacerbate inequality. With deliberate design and deployment, AI can be a powerful tool for amplifying human learning and helping bridge gaps. Students in our study not only learned more – they also seemed more engaged and motivated (as evidenced by increased voluntary attendance).

In closing, “post-solution” AI tutoring represents a promising approach to combining the scale and personalization of technology with the proven value of active student effort. As AI technology continues to evolve, we hope our findings encourage educators to embrace AI thoughtfully—leveraging its strengths (personalized feedback, endless patience, data-driven adaptation) while preserving the core of learning (challenge, effort, and human guidance). By doing so, we can move towards an educational paradigm where AI helps all students learn more efficiently, learn more deeply, and ultimately achieve their full potential.

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